PROPOSAL 2 · QUICK BRIEF

Offline Voice-Based Khata Entry in Hindi, Marathi and English for Kirana Stores

Research Methodology · what we are proposing, in plain words

In one line: can a free speech model running **offline on a cheap Android phone** correctly catch the **name, amount and credit/repayment** from a shopkeeper speaking mixed Hindi, Marathi and English, and is that faster than typing?

1 The problem

Typing is slow

At a busy counter, finding the customer and typing the amount takes time. It's harder still for people not comfortable typing.

No internet

Good speech recognition usually runs in the cloud. Shops lose network, and our app is offline-first.

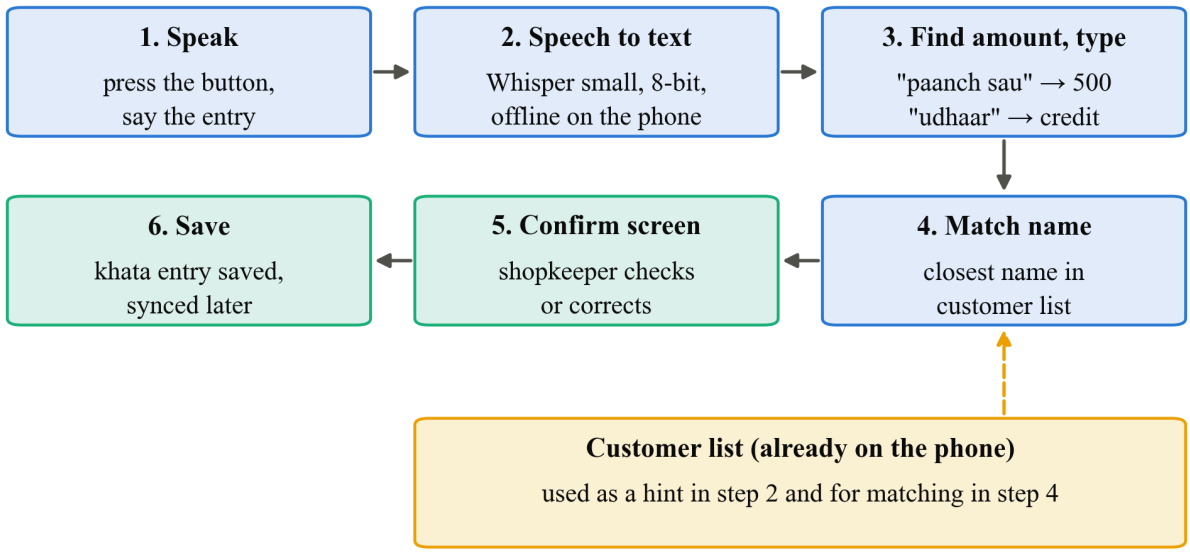
Mixed languages

People say "Ramesh ka 250 ka udhaar add karo", with Hindi, Marathi and English in one sentence.

For money, **only three details matter**: who, how much, credit or repayment. If "dhai sau" (250) is heard as "do sau" (200), the account is ₹50 off. So we judge by those details, not by word errors.

2 What we are proposing

A **"speak entry" button** in SmartOps that works fully offline:



Speak → phone turns speech into text offline → simple rules find amount and type → name matched to the shop's customer list → shopkeeper confirms → saved.

Example: "*Sunanda tai ne saade teen hazaar jama kele*" becomes **Sunanda Patil · ₹3,500 · Repayment**, and the shopkeeper taps Save.

3 What we compare

Baseline

Cloud service

Google speech-to-text. Needs internet. Our "best case" to compare with.

Offline

Whisper on the phone

Free OpenAI model, "small" size, compressed to 8-bit so it fits on a cheap phone.

Offline + hint

Whisper + name list

We give it the shop's customer names first, so rare names like "Sunanda" are heard right.

4 How we will test it

1. **Record about 360 clips:** 12 speakers (4 of us, 4 friends, 4 shopkeepers) × 30 short commands, read from cards or said in their own words. Names are made up, and we know the right answer for every clip.
2. **Run every clip through all 3 setups,** so it's the same speech for each, a fair comparison.
3. **Check:** is the amount right? the name? the whole entry? How long did the phone take?
4. **Usability check:** 8–10 people do 5 entries by voice and 5 by typing. We time them and they fill a 10-question SUS form. Half start with voice, half with typing.

₹

Amount correct

% of clips exactly right

Aa

Name correct

right customer picked

✓

Whole entry

name + amount + type all right (main measure)

s

Time on phone

fast enough on a cheap phone?

5 What we expect to show (hypotheses)

#	In plain words	Test	We are wrong if...
H1	Cloud gets more whole entries right than offline.	McNemar	no real difference, which would be good news for offline
H2	Giving the customer-name list improves name accuracy.	McNemar	the list doesn't help
H3	Voice entry is faster than typing.	Wilcoxon	typing is as fast or faster

McNemar, in one breath

Same clip, two setups. Ignore clips both got right or both got wrong, and check if one wins the disagreements far more than 50-50.

Wilcoxon, in one breath

For each person: voice time minus typing time. Check whether "voice faster" clearly outweighs "typing faster". No bell curve needed.

6 What we expect to find (expectations, not results)

- Cloud will be somewhat better. The question is *how big* the gap is. A small gap makes offline practical, especially with the confirm screen.
- Most mistakes will be in names and number words like "dhai", "saade", "paune". Mixed-language and noisy-shop clips will be harder.
- The name list should clearly improve name accuracy.
- Voice should be faster for most people. Some may prefer typing for privacy, since saying a customer's debt aloud can be awkward.

7 Plan and cost

12 weeks

Weeks 1–3: cards, forms, approval · 3–5: record clips · 3–7: build the voice feature · 6–8: run the setups · 8–9: usability test · 9–12: analysis and report.

About ₹11,000

Cheap Android phone ₹6,000, cloud usage ₹500, small thank-you payments to shopkeepers, printing and travel. Consent forms in Hindi, Marathi and English; clips stored as codes, not names.

8 Say it in 30 seconds

PITCH

"Shopkeepers find typing khata entries slow at a busy counter. Speaking is faster, but speech recognition usually needs internet. We're testing whether a free model, Whisper, running fully offline on a cheap Android phone can get the name, amount and credit-or-repayment right, even in mixed Hindi, Marathi and English. We'll record about 360 commands and compare offline against a cloud service, with and without a customer-name hint, judging by whether the whole entry is right. Then 8 to 10 people try voice versus typing. The output is an offline voice feature in SmartOps and honest numbers on how well it works."

9 Tough questions, quick answers

"Why not just use Google's cloud?"

We do, as the comparison. But the app must work without internet, so the real question is how close offline gets.

"360 clips is tiny."

It's small for training but fine for testing. Every clip goes through every setup, so comparisons are paired, and McNemar only needs the disagreements.

"Why only 8–10 people?"

Each person uses both methods, so we compare people with themselves. That's a normal size for a first usability check.

"Doesn't the confirm screen defeat the purpose?"

No. Checking one line and tapping Save is still much faster than typing, and it keeps money entries safe.

"Why rules, not ChatGPT, to pick out details?"

Rules work offline, are fast, and are easy to check and explain.

Full proposal: Research_Proposal_2_Voice_Khata.pdf · Detailed viva notes: explainer_voice.md